

This is unreleased documentation for Serverpod **Next** version.

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APPENDIX F

REFERENTIAL ACTIONS | SERVERPOD

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Referential actions

What happens to rows that reference a row you delete, or whose id changes? The `onUpdate` and `onDelete` properties on any relation answer that, and they map directly to the database's referential actions. In practice `onUpdate` rarely fires, since ids are stable.

Available referential actions

Action	Description
NoAction	If any constraint violation occurs, no action will be taken, and an error will be raised.
Restrict	If any referencing rows still exist when the constraint is checked, an error is raised.
SetDefault	The field will revert to its default value. Requires a default value on the field.
Cascade	Any action taken on the parent (update/delete) will be mirrored in the child.
SetNull	The field value is set to null. This action is permissible only if the field has been marked as optional.

Syntax

Use the following syntax to apply referential actions:

```
relation(onUpdate=<ACTION>, onDelete=<ACTION>)
```



Ask AI

Default values

If no referential actions are specified, both `onUpdate` and `onDelete` default to `NoAction`. This applies to [object relations](#) and [id relations](#) alike, so the two declarations below are equivalent to leaving the actions out:

```
parent: Model?, relation(onUpdate=NoAction, onDelete=NoAction)
parentId: int?, relation(parent=model_table, onUpdate=NoAction, onDelete=NoAction)
```

Deleting a parent row that is still referenced therefore fails with a foreign key violation unless you set `onDelete=Cascade` or `onDelete=SetNull` explicitly.

The order of `onUpdate` and `onDelete` in the relation does not matter.

Full example

```
class: Example
table: example
fields:
  parentId: int?, relation(parent=example, onUpdate=SetNull, onDelete=NoAction)
```

In the given example, if the `example` parent is updated, the `parentId` will be set to null. If the parent is deleted, no action will be taken for `parentId`.